

5G RAN

License Management Feature Parameter Description

Issue	Draft A
Date	2025-12-31



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History..... 1

2 About This Document..... 2

2.1 General Statements..... 2

2.2 Features in This Document..... 2

2.3 Differences..... 3

3 Introduction to Licenses and Application Scenarios of Licenses..... 5

3.1 Introduction to Licenses..... 5

3.2 Application Scenarios of Licenses..... 6

4 LKF Introduction..... 8

4.1 LKF Types..... 8

4.2 LKF States..... 9

4.2.1 Default State..... 10

4.2.2 Commissioning State..... 10

4.2.3 Normal State..... 10

4.2.4 Emergency State..... 10

4.2.5 Grace Period State..... 11

4.3 LKF Structure..... 11

4.3.1 Header..... 12

4.3.2 SaleInfo..... 14

4.3.3 KeyInfo..... 16

4.3.4 Footer..... 17

4.4 LKFs for Different Types of Base Stations..... 17

4.4.1 LKFs for Single-Mode Base Stations..... 17

4.4.2 LKFs for Multimode Base Stations..... 17

5 License Control Item Introduction..... 19

5.1 License Control Items..... 19

5.2 License Control Item Types..... 19

5.3 Validation and Invalidation Policies for a License Control Item..... 20

6 License Operations on the MAE-Access..... 22

6.1 Purchasing Licenses..... 24

6.2 Installing LKFs..... 24

6.2.1 Prechecking an LKF.....	24
6.2.2 Checking the Consistency Between an LKF to Be Installed and Data Configurations.....	24
6.2.3 Loading an LKF.....	26
6.2.4 Restoring an LKF.....	26
6.3 Modifying Configuration Data.....	27
6.3.1 Adjusting License Capacity.....	27
6.4 Maintaining LKFs.....	27
6.4.1 Synchronizing NE Information.....	27
6.4.2 Exporting NE Application Information.....	28
6.4.3 Exporting SBOM Statistics.....	28
6.4.4 Revoking an LKF.....	29
6.4.5 Checking Consistency Between Existing Licensed Values and Data Configurations.....	29
6.4.6 Backing Up an LKF.....	29
6.4.7 Obtaining a Backup LKF.....	30
7 License Operations on an LMT.....	31
7.1 Installing LKFs.....	32
7.1.1 Prechecking or Installing an LKF.....	32
7.1.2 Activating an LKF.....	32
7.1.3 Checking the Consistency Between Licensed Values and Data Configurations.....	33
7.2 Querying License Information.....	33
7.2.1 Querying LKF Information.....	33
7.2.2 Querying License Sales Information.....	33
7.3 Maintaining LKFs.....	33
7.3.1 Turning On the Emergency License State Switch.....	33
7.3.2 Setting the Switch Controlling Fixed-Term LKFs.....	34
7.4 Possible Issues.....	34
7.4.1 Faults Occurred During Installation.....	34
7.4.2 Alarms.....	38
7.4.3 Events.....	39
8 License Management Functions of Base Stations.....	40
8.1 LKF Precheck.....	40
8.2 LKF Verification.....	40
8.3 Consistency Check Between Licensed Values and Data Configurations.....	41
8.4 LKF Revocation.....	42
8.5 License Overload.....	42
8.6 LKF Backup and Restoration.....	43
8.7 Invalidating Fixed-term License Items.....	44
9 Parameters.....	45
10 Counters.....	47
11 Glossary.....	48

12 Reference Documents.....

49

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-12-14).

- Technical changes

None

- Editorial changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include chapter/section reorganization and rewording of information.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-010025	Basic O&M Package	3 Introduction to Licenses and Application Scenarios of Licenses 4 LKF Introduction 5 License Control Item Introduction 6 License Operations on the MAE-Access 7 License Operations on an LMT 8 License Management Functions of Base Stations

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
License Management	None	3 Introduction to Licenses and Application Scenarios of Licenses 4 LKF Introduction 5 License Control Item Introduction 6 License Operations on the MAE-Access 7 License Operations on an LMT 8 License Management Functions of Base Stations

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 as high frequency bands. For details about FR1 and FR2, see.

Function Name	Difference	Chapter/Section
License Management	Supported in both high and low frequency bands, with the following differences: A high-frequency feature with the sales unit of per Cell takes effect in all activated cells in a sector only after the license controlling this feature is obtained for such cells in the sector. If license insufficiency occurs, then no cells in the sector consume this license, and the feature controlled by this license does not take effect in any cell in the sector.	3 Introduction to Licenses and Application Scenarios of Licenses 4 LKF Introduction 5 License Control Item Introduction 6 License Operations on the MAE-Access 7 License Operations on an LMT 8 License Management Functions of Base Stations

3 Introduction to Licenses and Application Scenarios of Licenses

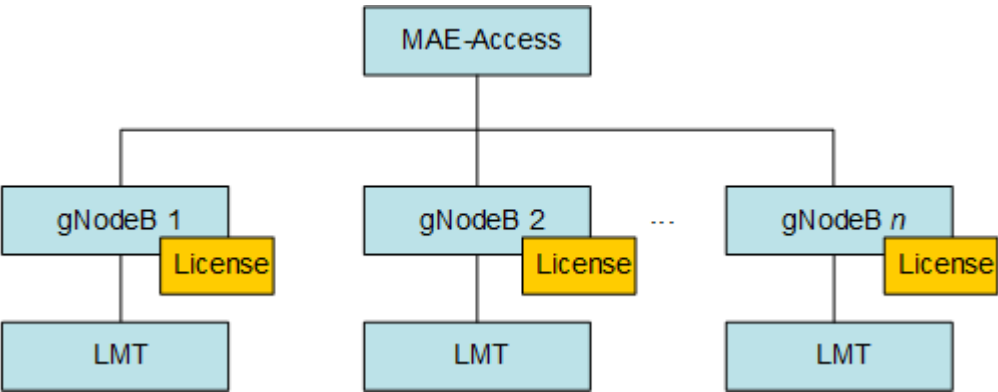
3.1 Introduction to Licenses

Licenses are contracts that specify the features, releases, capacities, validity periods, and usage scope of an operator's purchased products. All license information is carried by license key files (LKFs).

License management ensures that the system properly operates with licensed features, versions, capacities, and usage scope during licensed periods. As shown in [Figure 3-1](#), operators can remotely download an LKF to a gNodeB and manage as well as query LKF content on a local maintenance terminal (LMT) or MAE-Access client.

The LKF of a gNodeB is stored on the main control board.

Figure 3-1 License management architecture



NOTE

Unless otherwise specified, the NEs mentioned in the following sections refer to gNodeBs.

3.2 Application Scenarios of Licenses

The entity of a license is an LKF. An LKF is required in the following scenarios.

New Base Station Deployment

New deployment involves the engineering phase and maintenance phase. At the engineering phase, a commissioning LKF with a 255-day validity period and **ESN** set to **ANY** is installed on the base station for commissioning. After a purchase order is placed, Huawei engineers apply for commercial LKFs based on the network plan.

Operations Related to Commissioning LKFs

During new deployment, commissioning LKFs must be installed on NEs for commissioning, and commercial LKFs can be installed after engineering commissioning is completed. This is because commercial LKFs of the NEs are not available due to uncertain network plans. Before installing base stations, contact Huawei engineers to obtain a commissioning LKF with a 255-day validity period and **ESN** set to **ANY**. The commissioning LKF can be installed on any base station to facilitate deployment.

Operations Related to Commercial LKFs

Commissioning LKFs are valid for only a limited period of time. When newly deployed base stations operate stably, commercial LKFs must be created for the base stations based on the network plan and installed on the base stations by Huawei engineers. Engineers may create commercial LKFs in a batch and install them on the corresponding base stations in a single operation. For details, see [6 License Operations on the MAE-Access](#).

Version Upgrade or Capacity Expansion

When R version upgrades or capacity expansions are required for product software to meet requirements for increasing traffic volume or new services, new LKFs must be applied for to replace the existing ones.

Before Huawei engineers perform an upgrade or capacity expansion for product software, the current LKF and ESN information of NEs needs to be obtained. For details about how to obtain the NE information, see [6 License Operations on the MAE-Access](#).

Licensed Capacity Adjustment

If the network plan is adjusted, commercial LKFs on NEs must be re-created with adjusted licensed capacities for the NEs by Huawei engineers.

Before re-creating an LKF for an NE, obtain the current LKF information, ESN, and LKF revocation code of the NE by referring to [6 License Operations on the MAE-Access](#).

 NOTE

If a commercially permanent & fixed-term LKF is used, capacities can be adjusted only for permanent license control items, and capacities for fixed-term license control items remain unchanged.

ESN Change

When the ESN of an NE changes, the original LKF is unusable, and a new LKF is required. First, Huawei engineers revoke the original LKF by referring to [6 License Operations on the MAE-Access](#) and obtain the revocation code. Then, based on the revocation code and new ESN, Huawei engineers apply for a new LKF and install it on the main control board.

 NOTE

If the BBU backplane of a base station changes, the ESN will change. In this case, you need to apply for a new LKF. If the main control board of a base station changes, the ESN will not change, and you do not need to apply for a new LKF.

4 LKF Introduction

4.1 LKF Types

LKFs are classified into commercial LKF, commercially permanent & fixed-term LKF, and commissioning LKF based on application scenarios.

Commercial LKF

A commercial LKF is created on the basis of a business contract. In principle, a license certificate is issued in electronic form to the customer. The authorization type (**LicType**) for a commercial LKF is **COMM**.

License control items in a commercial LKF are either all permanent ones with the expiration date being **PERMANENT** or all those having the same fixed expiration date and a default grace period of 60 days.

Commercially Permanent & Fixed-term LKF

A commercially permanent & fixed-term LKF is used when a network needs to be frequently adjusted and a license trial is required before purchase. The authorization type (**LicType**) for a commercially permanent & fixed-term LKF is **COMM**.

A commercially permanent & fixed-term LKF comprises permanent license control items and license control items having a fixed expiration date. The expiration date is **PERMANENT** for permanent license control items.

Commissioning LKF

A commissioning LKF is temporarily used to verify software functions during project implementation. In principle, a license certificate is issued in electronic form to the customer.

The authorization type (**LicType**) is **DEMO**, and all license control items in a commissioning LKF have the same expiration date.

Commissioning LKFs are further classified into the following subcategories:

- Commissioning LKFs with a 255-day validity period and **ESN** set to **ANY**: There are no restrictions on the number of times that commissioning LKFs of this type can be loaded within the 255-day time period. This type of LKF can be used for a maximum of 255 days on a single gNodeB within one software release.
- Commissioning LKFs with a 3-day validity period and **ESN** set to **ANY**: There are no restrictions on the number of times that commissioning LKFs of this type can be loaded within the 3-day time period. The use duration of this type of commissioning LKF is not added to the total number of days that commissioning LKFs with **ESN** set to **ANY** are used. Therefore, commissioning LKFs of this type are not subject to the restriction that commissioning LKFs with **ESN** set to **ANY** cannot be used for more than 255 days within one software release. This type of LKFs is used only for fault rectification.
- Commissioning LKFs with a 14-day validity period and **ESN** set to a value other than **ANY**: There are no restrictions on the number of times that commissioning LKFs of this type can be loaded within the 14-day time period. This type of LKF applies only to urgent fault recovery scenarios.

NOTE

In the **DSP LICINFO** command output, the value of **Remaining Days for Commissioning License with ESN of ANY** is not fixed. It is calculated by deducting the number of days in which such an LKF has been used from the expiration time of the last successfully loaded commissioning LKF. The number of days in which such an LKF has been used is accumulated from zero in each version. When a new version is used, the value is accumulated from zero again.

4.2 LKF States

An LKF may be in one of the following states: default, commissioning, normal, emergency, and grace period. The LKF states are as follows:

- Default: If no valid LKF has been installed on an NE during new site deployment, the NE's LKF is in the default state.
- Commissioning: If a commissioning LKF has been installed on an NE during engineering implementation, the NE's LKF is in the commissioning state.
- Normal: If a commercial or commercially permanent & fixed-term LKF has been installed on an NE during the acceptance stage, the NE's LKF is in the normal state.
- Emergency: In case of an emergency (such as an earthquake) during the maintenance stage, if the emergency license has been activated by running the **SET LICENSECTRL** command, the NE's LKF is in the emergency state.
- Grace period: When an NE is operating, the NE's LKF is in the grace period state if any of the following conditions applies:
 - An **ESN** or **Version** check on an LKF fails.
 - A commissioning LKF expires.
 - A commercial LKF or a commercially permanent & fixed-term LKF is revoked.
 - In a commercially permanent & fixed-term LKF, the fixed-term license control items expire.

The following sections describe the LKF states in detail.

4.2.1 Default State

The LKF of an operating NE enters the default state if either of the following conditions applies:

- No valid LKF is installed on the NE.
- The NE's LKF has been in the grace period for 60 days.

In the default state, an activated cell can continue to use enabled features until the cell is reset. This ensures function continuity. After the cell is reset, the NE operates at the default capacity of license control items, and all optional function items are disabled.

When the NE's LKF enters the default state, it reports ALM-26818 No License Running in System.

4.2.2 Commissioning State

The LKF of an operating NE enters the commissioning state when a commissioning LKF is installed on the NE. In a commissioning LKF, resource items are set to the maximum values and optional function items are all enabled.

For a fixed-term license control item in a commissioning LKF, ALM-26823 License Feature About to Expire is reported when the period between the current time and the expiration date of the fixed-term license control item is detected to be less than or equal to the related threshold.

After a commissioning LKF expires, the NE's LKF status changes from the commissioning state to the grace period state. An NE's LKF can stay in the grace period for up to 60 days. After 60 days elapse, the LKF enters the default state.

4.2.3 Normal State

The LKF of an operating NE enters the normal state when a commercial or commercially permanent & fixed-term LKF is installed on the NE. In the normal state, the NE operates within the licensed values defined in the LKF.

NOTICE

For a fixed-term license control item in an LKF, ALM-26823 License Feature About to Expire is reported when the period between the current time and the expiration date of the fixed-term license control item is detected to be less than or equal to the related threshold. However, for a commercially permanent & fixed-term LKF, if the fixed-term license switch is turned off, ALM-26823 License Feature About to Expire is not reported.

If license control items in the fixed-term feature segments of a commercially permanent & fixed-term LKF expire, the NE reports ALM-26815 Licensed Feature Entering Keep-Alive Period. After 60 days elapse, these license control items become invalid, and the NE reports ALM-26816 Licensed Feature Unusable.

4.2.4 Emergency State

The emergency switch (***SWITCH***) can be set to **ON** for a gNodeB on special occasions (such as earthquakes) to activate the emergency license. After the

emergency license is activated, the gNodeB serves the maximum number of UEs, instead of being confined to the licensed limits, preventing performance deterioration. After the emergency license is activated for a gNodeB originally using an LKF in the default state, the gNodeB operates with the maximum allowable capabilities in terms of the number of UEs and other license control items. Once the emergency license is activated, the emergency state has a duration of seven days, and it can be triggered a maximum of three times for each gNodeB in the same release. When the gNodeB's LKF enters or exits the emergency state, it reports an event:

- Entering the state: EVT-26820 License Emergency Status Activated
- Exiting the state: EVT-26821 License Emergency Status Ceased

The **LST LICENSECTRL** command can be used to query **Emergency Switch State**, **Remaining Emergency Times**, and **Remaining Emergency Days**.

NOTE

For a fixed-term license control item in an emergency-state LKF, ALM-26823 License Feature About to Expire is reported when the period between the current time and the expiration date of the fixed-term license control item is detected to be less than or equal to the related threshold. However, for a commercially permanent & fixed-term LKF, if the fixed-term license switch is turned off, ALM-26823 License Feature About to Expire is not reported.

4.2.5 Grace Period State

The LKF of an operating NE enters the grace period state when (1) an **ESN** or **Version** check on its LKF fails, (2) its commissioning LKF expires, (3) its commercial LKF or commercially permanent & fixed-term LKF is revoked, or (4) the fixed-term license control items expire in a commercially permanent & fixed-term LKF.

In the grace period state, the NE can operate at the licensed values for up to 60 days without impacting services. After 60 days, the NE's LKF enters the default state.

When the NE's LKF enters the grace period state, it reports ALM-26817 License on Trial. The reporting of ALM-26817 License on Trial does not affect ongoing services. If O&M engineers manually clear or mask this alarm and do not install a commercial LKF promptly, network accidents may arise. To avoid this situation, the NE will check and report the alarm the next day after the alarm is cleared, and O&M engineers cannot mask the alarm.

4.3 LKF Structure

LKFs are saved as .xml files. [Table 4-1](#) describes the information contained in an LKF. LKFs contain only the purchased features and capacities.

Table 4-1 LKF structure

Segment	Content
Header	General information that describes an LKF, including the GeneralInfo , CustomerInfo , and NodeInfo sections
SaleInfo	Customer-oriented sales information, including the product information and sales items
KeyInfo	Software control information
Footer	LKF signature

The use of LKFs involves the following value types:

- **Licensed value:** indicates the allocated value and refers to the number of hardware or feature licenses purchased by a customer. If the licensed value is smaller than the number of hardware devices or features, those that exceed the licensed value cannot be used, which may affect service functions or performance.
- **Configured value:** refers to the number of hardware or feature licenses configured on an NE for a customer. The configured value indicates license requirements. For example, if the sales unit of a feature license is per Cell, the configured value is the number of cells for which the feature switch is turned on. The configured value is used to evaluate the number of license units required and is used as a reference for the licensed value. It is not affected by actual hardware or feature activation.
- **Actually used value:** refers to the number of hardware or feature licenses actually used by a customer. For example, if the sales unit of a feature license is per Cell, the actually used value for a cell is 0 when the cell is not activated.

4.3.1 Header

GeneralInfo

GeneralInfo indicates the general information about an LKF. [Table 4-2](#) describes the fields in the **GeneralInfo** section.

Table 4-2 Fields in the GeneralInfo section

Field	Field Description
CopyRight	Copyright, which is always "Huawei Technologies Co., Ltd. All rights reserved."
LSN	Serial number of the LKF
LicType	Type of the LKF

Field	Field Description
GraceDay	Number of days in a grace period, during which licensed capacities and features are still in effect, after the LKF expires
Creator	Company that created the LKF, which is always "Huawei Technologies Co., Ltd."
Issuer	Name of the user who creates the LKF
CreateTime	Time when the LKF was created

CustomerInfo

CustomerInfo indicates the customer information about an LKF. [Table 4-3](#) describes the fields in the **CustomerInfo** section.

Table 4-3 Fields in the CustomerInfo section

Field	Field Description
Customer	Operator name
Country	Country where the operator is located
Office	City where the operator is located
MCC	Mobile country code (MCC)
MNC	Mobile network code (MNC)

NOTE

The MCC and MNC fields in the CustomerInfo section are unique to commissioning license files.

NodeInfo

NodeInfo indicates the NE information about an LKF. [Table 4-4](#) describes the fields in the **NodeInfo** section.

Table 4-4 Fields in the NodeInfo section

Field	Field Description
Node	Name of the NE that the LKF applies to
Des	Detailed description of the NE

Field	Field Description
Sequence	Sequence number. A value x indicates that this LKF is the x th LKF for the NE. The value increases each time the LKF undergoes an upgrade, capacity expansion, reconfiguration, equipment serial number (ESN) change, or issuer change.
LockModel	Locking mode, which consists of a maximum of 16 bytes. This field can be set to NODELOCK ^a (default value) or DUALLOCK ^b .
ESN	ESN of the NE
a: NODELOCK : The LKF can only be used on a specified NE. b: DUALLOCK : The LKF can be used on several specified NEs.	

4.3.2 SaleInfo

OfferingProduct

OfferingProduct indicates the product information about an LKF. [Table 4-5](#) describes the fields in the **OfferingProduct** section.

Table 4-5 Fields in the OfferingProduct section

Field	Field Description
name	Product name
version	Product version

SaleItem

SaleItem indicates the sales item information about an LKF. Each **SaleItem** section has a **name** attribute, which indicates the sales bill of material (SBOM) part number of the sales item. The **SaleItem** section contains only the purchased features and capacities.

[Table 4-6](#) describes the fields in each **SaleItem** section.

Table 4-6 Fields in the SaleItem section

Field	Field Description	Attribute	Description
Value	Purchased units of the sales item	validDate	Expiration date in the <i>YYYY-MM-DD</i> format. This attribute has a value of PERMANENT if the item never expires.
DesChs/ DesEng	Chinese or English description of the sales item	unit	Sales unit
CapacityKey	Resource item. Table 4-7 describes its subordinate field. One SaleItem section may contain more than one CapacityKey field.	name	Name of a license control item
FeatureKey	Function item. Table 4-7 describes its subordinate field. One SaleItem section may contain more than one FeatureKey field.	name	Name of a license control item
Note: If both permanent and fixed-term license control items exist, the section may include more than one Value field with different validDate attributes.			

Table 4-7 Field under CapacityKey and FeatureKey

Field	Field Description	Attribute	Description
Value	Licensed value	validDate	Expiration date in the <i>YYYY-MM-DD</i> format. This attribute has a value of PERMANENT if the item never expires.
Note: If both permanent and fixed-term license control items exist, the section may include more than one Value field with different validDate attributes.			

A **CapacityKey** (or **FeatureKey**) field may have different **Value** settings in the **SaleInfo** and **KeyInfo** segments. **Value** in the **SaleInfo** segment indicates sales information, and that in the **KeyInfo** segment indicates software control

information. The **Value** settings in the two segments can be interconverted and may be different.

4.3.3 KeyInfo

The **KeyInfo** segment includes the fields described in [Table 4-8](#).

Table 4-8 Fields in the **KeyInfo** segment

Field	Field Description
productName	Product name
productVersion	Product version

The **KeyInfo** segment contains at least one **CapacityKey** or **FeatureKey** field. [Table 4-9](#) describes the fields in the **KeyInfo** segment.

Table 4-9 Fields in the KeyInfo segment

Field	Field Description	Attribute	Description
CapacityKey	Resource item. Table 4-10 describes its subordinate field.	name	Name of a license control item
		group	Group that the license control item belongs to
FeatureKey	Function item. Table 4-10 describes its subordinate field.	name	Name of a license control item
		group	Group that the license control item belongs to

Table 4-10 Field under CapacityKey and FeatureKey

Field	Field Description	Attribute	Description
Value	Licensed value	validDate	Expiration date in the <i>YYYY-MM-DD</i> format. This attribute has a value of PERMANENT if the item never expires.
Note: If both permanent and fixed-term license control items exist, the section may include more than one Value field with different validDate attributes.			

4.3.4 Footer

Table 4-11 describes the fields in the **Footer** segment.

Table 4-11 Fields in the **Footer** segment

Field	Field Description	Attribute	Description
CommonSign	Common signature	method	ID of the encryption algorithm
ProductSign	Customized signature	method	ID of the encryption algorithm NOTE Only RSA3072 can be used as the encryption algorithm.

 NOTE

In V100R020C00 and later versions, only RSA3072 is supported as the digital signature encryption algorithm of standalone LTE and NR license files.

4.4 LKFs for Different Types of Base Stations

Base stations can be classified into single-mode and multimode base stations based on the number of modes that they support.

4.4.1 LKFs for Single-Mode Base Stations

A single-mode base station refers to the NR gNodeB.

The LKF is stored on the gNodeB, and the gNodeB implements functions such as loading, controlling, querying, and verifying the LKF.

4.4.2 LKFs for Multimode Base Stations

Multimode base stations can work in GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, as well as NR, and they are classified into co-MPT and separate-MPT base stations.

- A co-MPT base station has only one OMCH with the MAE, and it is managed and maintained as one logical NE (BTS3900/BTS5900) on the MAE-Access.
- A separate-MPT base station has multiple OMCHs with the MAE, and it is managed and maintained as a number of logical NEs on the MAE-Access.

There is no dedicated LKF for a multimode base station (co-MPT or separate-MPT). The GBTS, eGBTS, NodeB, eNodeB, or gNodeB in a multimode base station has a separate LKF and manages its license independently. The details are as follows:

- GBTS: The GBTS is managed as a board of the base station controller (BSC), and its license is also managed by the BSC.

- eGBTS: The eGBTS's LKF is stored on the MAE-Access and shared by all other eGBTSs under the same BSC.
- NodeB: The NodeB's LKF is stored on the MAE-Access. NodeB license management is the same as that of a single-mode NodeB.
- eNodeB: The eNodeB can work in LTE FDD, LTE TDD, or NB-IoT; however, it uses the same LKF with different license control items for LTE FDD, LTE TDD, and NB-IoT. The LKF is stored on the multimode base station and license management is the same as that of a single-mode eNodeB (LTE FDD, LTE TDD, or NB-IoT).
- gNodeB: The LKF is stored on the multimode base station and license management is the same as that of a single-mode gNodeB.

5 License Control Item Introduction

5.1 License Control Items

License control items are key components in an LKF, and they are displayed as models in the LKF. After an LKF is installed and functions properly, license control items that are in use can be queried by running the **DSP LICINFO** command, and their descriptions will be displayed in the command output. The model and description of a license control item are described as follows:

- **Model**
A license control item is displayed as a model in an LKF and consists of uppercase letters and digits. A model uniquely identifies a license control item.
- **Description**
The description of a license control item is similar to its feature or capacity name. For details about the mapping between license control item descriptions and feature or capacity names, see *License Control Item Lists*.

NOTE

License control item descriptions are displayed in the **Description** column in the **DSP LICINFO** command output.

5.2 License Control Item Types

An LKF is used on multimode and single-mode base stations. License control items are classified into the following two types:

- **Capacity items**
Operators can purchase license units for the level of capacity that they require without spending money for capacity that they will not use.
- **Feature items**
Feature items cover basic and optional software features. Basic software features are usually assembled into a basic package for sale. The basic package is mandatory.

For details about license meaning, feature mapping, RAT, and licensing principles, see *License Control Item Lists*.

5.3 Validation and Invalidation Policies for a License Control Item

When the licensed value of a license control item changes, for example, due to loading of a new LKF, the policies for the license control item to validate or invalidate the capacity or feature are as follows:

- When the licensed value of a license control item increases, the capacity or feature controlled by the license control item takes effect immediately.
- When the licensed value of a license control item decreases and the number of licenses is insufficient, the capacity or feature controlled by the license control item is not invalidated immediately.
 - The capacity or feature controlled by the license control item is invalidated only when the cell is reactivated, the feature switch is turned on again, or UEs initiate the access request again.
 - If a feature controlled by a license control item contains multiple sub-features and each sub-feature is controlled by a specific switch, changing the setting of any sub-feature switch will trigger the release of the license control item, which leads to a decrease in the used value. In this case, sub-features that do not take effect are prohibited from taking effect. If the settings of all sub-feature switches remain unchanged, the sub-features that have taken effect remain in effect until the conditions for invalidating them are met.

In independent SUL deployment mode, SUL cells do not consume the capacity items listed in the following table.

Model	Description
NR0S00RFDM00	Multimode License for RF Module (NR)
NR0S5000MMRR	Multimode License for 5000 Series RF Module (NR)
NR0S5000FMRR	First Mode License for 5000 Series RF Module (NR)
NR0SMBRFMD00	Multi-band License for 5000 Series Multi-band RF Module (NR)
NR0SMMRFMD00	Massive MIMO Multimode License for 5000 Series RF Module (NR)
NR0SMMMBMD00	Massive MIMO Multi-band License for 5000 Series Multi-band RF Module (NR)
NR0SMMMBMB00	Multi-Mode Multi-Band License for Multi-Band RF Module (NR)
NR0SFMFDMD00	First Mode First Band License for Multi-Band RF Module (NR)

Model	Description
NR0SFSTMOD00	First Mode License for 5000 Series LampSite pRRU(NR)
NR0SMUTMOD00	Multimode License for 5000 Series LampSite pRRU(NR)

6 License Operations on the MAE-Access

Table 6-1 lists the license operations on the MAE-Access. For details, see **NE License Management** under **Operation and Maintenance > MAE-Access Operation and Maintenance > RAN Management** in *iMaster MAE Product Documentation*.

Table 6-1 License operations on the MAE-Access

Section	Description	Functions on the Base Station Side
6.4.1 Synchronizing NE Information	Describes how to synchronize ESN and license information of NEs to the MAE-Access.	-
6.2.1 Prechecking an LKF	Describes how to precheck LKFs to be installed.	8.1 LKF Precheck
6.2.3 Loading an LKF	Describes how to install commercial or commissioning LKFs on NEs.	8.2 LKF Verification
6.4.4 Revoking an LKF	Describes how to revoke a commercial LKF and how to export the revocation information to the MAE-Access.	8.4 LKF Revocation
6.3.1 Adjusting License Capacity	Describes how to adjust the license capacity of NEs using the MAE-Access.	-

Section	Description	Functions on the Base Station Side
6.4.5 Checking Consistency Between Existing Licensed Values and Data Configurations	Describes how to check consistency between existing licensed values and data configurations and how to export the result to the MAE-Access.	8.3 Consistency Check Between Licensed Values and Data Configurations
6.2.2 Checking the Consistency Between an LKF to Be Installed and Data Configurations	Describes how to check consistency between the LKF to be installed and data configurations and how to export the result to the MAE-Access.	8.3 Consistency Check Between Licensed Values and Data Configurations
6.4.6 Backing Up an LKF	Describes how to back up the LKFs of base stations on the MAE-Access server.	8.6 LKF Backup and Restoration
6.4.7 Obtaining a Backup LKF	Describes how to retrieve the backup LKFs of base stations to the MAE-Access from the MAE-Access server.	8.6 LKF Backup and Restoration
6.2.4 Restoring an LKF	Describes how to reload the LKFs, which are backed up on the MAE-Access server or locally on the MAE-Access, on NEs.	8.6 LKF Backup and Restoration
6.4.3 Exporting SBOM Statistics	Describes how to export SBOM statistics of NEs to the MAE-Access.	-

NOTE

When performing license operations on the MAE-Access, select **5G** for the product type in the displayed **Stand-Alone License Management** window.

6.1 Purchasing Licenses

Operators can purchase licenses to select network functions and capacities flexibly at specific stages. This helps increase return on investment and reduces network deployment costs.

6.2 Installing LKFs

6.2.1 Prechecking an LKF

To precheck an LKF, perform the following steps:

- Step 1** Choose **License > NE License Management > Stand-Alone**.
- Step 2** In the left navigation pane, choose **Stand-Alone License**.
- Step 3** Select an NE tab.
- Step 4** Click **More**, and select **Check License**.
- Step 5** On the **Check License** page, set **Compare Mode** to **Check license**.
- Step 6** Select an LKF, and click the **Upload** button to upload the LKF.

The following uses an LKF in TXT, DAT, or XML format as an example:

1. Select an LKF in TXT, DAT, or XML format. In the **Check License** dialog box, click **Next**.
 2. In the **NE Selection** phase, select the NEs for which the task is to be executed.
 3. Click **Next**. View the check task execution in the **Task Execution** phase.
 4. After the task execution is successful, select the task, and click **Load**.
 5. In the displayed dialog box, select **Selected data**, **Data that passes the check in the selected data**, or **All data that passes the check**, and perform operations as prompted.
- Step 7** Click **Close**.

----End

6.2.2 Checking the Consistency Between an LKF to Be Installed and Data Configurations

Context

Consistency check between licensed values and data configurations can be used in the following scenarios:

- LKF installation
If a license control item has the corresponding configuration data but the licensed value in the LKF is less than the configured value, the LMT returns a minor error message while the MAE-Access returns a major error message

during LKF installation. If the LKF is forcibly installed, the NE reports ALM-26819 Data Configuration Exceeding Licensed Limit and the configured data beyond the licensed value does not take effect. For details, see [7.4.1 Faults Occurred During Installation](#).

NOTE

If the license control item gNodeB Basic Software is unavailable and the licensed value in the newly installed LKF is less than the corresponding configured value, the following are true:

- If the licensed value in the new LKF is greater than or equal to that in the current LKF (the licensed value in the current LKF is 0 when the NE's LKF enters the default state), the LMT displays a minor error message and the MAE-Access displays a major error message during LKF installation. If the LKF is forcibly installed, the NE reports ALM-26819 Data Configuration Exceeding Licensed Limit and the configured data beyond the licensed value does not take effect.
- If the licensed value in the new LKF is less than that in the current LKF (the licensed value in the current LKF is 0 when the NE's LKF enters the default state), the LKF installation fails.
- **Data configuration**
An NE compares data configurations with licensed values when it attempts to make the data configurations take effect, rather than when the data is configured. Data can be written into the database when data configuration constraints are fulfilled. When the NE attempts to make the data configurations take effect, a failure message is displayed if the NE finds that the configured data is greater than the licensed values. In addition, the NE reports ALM-26819 Data Configuration Exceeding Licensed Limit.
- **License maintenance before and after LKF installation**
Before a new LKF is installed on an NE, related consistency check can be performed to verify whether the LKF meets the NE configuration requirements. For details about how to perform the check, see [6 License Operations on the MAE-Access](#). Consistency check can also be performed when ALM-26819 Data Configuration Exceeding Licensed Limit is reported because the delivered configuration data exceeds the licensed values after an LKF is installed on an NE. In this situation, consistency between the LKF and NE data configurations can be checked. For details about how to perform the check, see [6 License Operations on the MAE-Access](#).

Procedure

To check the consistency between LKFs to be installed and data configurations, perform the following steps:

- Step 1** Choose **License > NE License Management > Stand-Alone**.
- Step 2** In the left navigation pane, choose **Stand-Alone License**.
- Step 3** Select an NE tab.
- Step 4** Click **More**, and select **Check License**.
- Step 5** On the **Check License** page, set **Compare Mode** to **Compare configuration data**.
- Step 6** Select an LKF, and click the **Upload** button to upload the LKF.

The following uses an LKF in TXT, DAT, or XML format as an example:

1. Select an LKF in TXT, DAT, or XML format. In the **Check License** dialog box, click **Next**.
2. In the **NE Selection** phase, select the NEs for which the task is to be executed.
3. Click **Next**. View the check task execution in the **Task Execution** phase.
4. After the task execution is successful, select the task, and click **Load**.
5. In the displayed dialog box, select **Selected data**, **Data that passes the check in the selected data**, or **All data that passes the check**, and perform operations as prompted.

Step 7 Click **Close**.

----End

6.2.3 Loading an LKF

To load an LKF, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, and select **Load License File**.

Step 5 Select an LKF, and click the **Upload** button to upload the LKF.

The following uses an LKF in TXT, DAT, or XML format as an example:

1. Select an LKF in TXT, DAT, or XML format. In the **Check License** dialog box, click **Next**.
2. In the **NE Selection** phase, select the NEs for which the task is to be executed.
3. Click **Next**. View the check task execution in the **Task Execution** phase.
4. After the task execution is successful, select the task, and click **Load**.
5. In the displayed dialog box, select **Selected data**, **Data that passes the check in the selected data**, or **All data that passes the check**, and perform operations as prompted.

Step 6 Click **Close**.

----End

6.2.4 Restoring an LKF

To restore an LKF, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Restore License File**. In the displayed dialog box, set task information.

Step 5 Select **Active** or **Inactive** in the **License Type** area in the **Restore License File** dialog box.

- If you select **Active**, the MAE-Access uses an active backup file to restore the LKF. That is, the MAE-Access uses the backup file under **/export/home/ossshare/MAE/NeLicenseService/var/standaloneNeLicense/active/<Product name>** in the server to restore the LKF.
- If you select **Inactive**, the MAE-Access uses an inactive backup file to restore the LKF. That is, the MAE-Access uses the backup file under **/export/home/ossshare/MAE/NeLicenseService/var/standaloneNeLicense/inactive/<Product name>** in the server to restore the LKF.

Step 6 Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.

Step 7 Click **Next**. View the check task execution in the **Task Execution** phase.

Step 8 Click **Close**.

----End

6.3 Modifying Configuration Data

6.3.1 Adjusting License Capacity

To adjust the license capacity, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Adjust License Capacity**. In the displayed dialog box, set task information.

Step 5 Select the file type of the exported result. The file can be saved in XLSX, XLS, CSV, or XML format. In an XLS or XLSX file, a cell supports a maximum of 32,767 characters. Click **Next**.

Step 6 Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.

Step 7 Click **Next**. View the check task execution in the **Task Execution** phase. After the task is successfully executed, the file is automatically exported.

Step 8 In the **Stand-Alone License** panel, choose **More > Load License File** to load the new LKF onto the NE. For details, see [6.2.3 Loading an LKF](#).

----End

6.4 Maintaining LKFs

6.4.1 Synchronizing NE Information

To synchronize NE information, perform the following steps:

- Step 1** Choose **License > NE License Management > Stand-Alone**.
- Step 2** In the left navigation pane, choose **Stand-Alone License**.
- Step 3** Select an NE tab.
- Step 4** Click **Synchronizing NE Information**. In the displayed dialog box, set task information.
- Step 5** Click **Close**.
- End

6.4.2 Exporting NE Application Information

To export NE application information, perform the following steps:

- Step 1** Choose **License > NE License Management > Stand-Alone**.
- Step 2** In the left navigation pane, choose **Stand-Alone License**.
- Step 3** Select an NE tab.
- Step 4** Click **More**, select **Export NE Inter-Site Adjustment Reference**. In the displayed dialog box, set task information.
- Step 5** In the **Basic Information** phase, set basic parameters.
- Step 6** Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.
- Step 7** Click **Next**. View the check task execution in the **Task Execution** phase.
- Step 8** Click **Close**.
- Step 9** Send the exported NE application information to Huawei technical support engineers for them to apply for a stand-alone LKF.
- End

6.4.3 Exporting SBOM Statistics

To export SBOM statistics, perform the following steps:

- Step 1** Choose **License > NE License Management > Stand-Alone**.
- Step 2** In the left navigation pane, choose **Stand-Alone License**.
- Step 3** Select an NE tab.
- Step 4** Click **More**, select **Export SBOM Statistics**. In the displayed dialog box, set task information.
- Step 5** In the **Basic Information** phase, set basic parameters.
- Step 6** Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.
- Step 7** Click **Next**. View the check task execution in the **Task Execution** phase.

Step 8 Click **Close**.

----End

6.4.4 Revoking an LKF

To revoke an LKF, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Revoke License**. In the displayed dialog box, set task information.

Step 5 In the **Basic Information** phase, set basic parameters.

Step 6 Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.

Step 7 Click **Next**. View the check task execution in the **Task Execution** phase.

Step 8 Click **Close**.

----End

6.4.5 Checking Consistency Between Existing Licensed Values and Data Configurations

To check the consistency between existing licensed values and data configurations, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Check License**. In the displayed dialog box, set task information.

Step 5 In the **Basic Information** phase, set basic parameters. Set **Compare Mode** to **Check configuration data** and **Select License File** to **Existing License File**.

Step 6 Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.

Step 7 Click **Next**. View the check task execution in the **Task Execution** phase.

Step 8 Click **Close**.

----End

6.4.6 Backing Up an LKF

To back up an LKF, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Back Up License File**. In the displayed dialog box, set task information.

Step 5 Click **Next**. In the **NE Selection** phase, select the NEs for which the task is to be executed.

Step 6 Click **Next**. View the check task execution in the **Task Execution** phase.

Step 7 Click **Close**.

----End

6.4.7 Obtaining a Backup LKF

To obtain a backup LKF, perform the following steps:

Step 1 Choose **License > NE License Management > Stand-Alone**.

Step 2 In the left navigation pane, choose **Stand-Alone License**.

Step 3 Select an NE tab.

Step 4 Click **More**, select **Obtain Backup License File**. In the displayed dialog box, set task information.

Step 5 Click **OK**.

Step 6 Export the stand-alone backup license file.

----End

7 License Operations on an LMT

Table 7-1 lists the license operations on an LMT.

Table 7-1 License Operations on an LMT

To...	Run...	Description
7.1.1 Prechecking or Installing an LKF	INS LICENSE	This command is used to precheck or install an LKF.
7.1.2 Activating an LKF	ACT LICENSE	This command is used to activate an LKF on an NE.
Upload an LKF	ULD LICENSE	This command is used to upload an LKF for NE-specific LKF backup.
Revoke an LKF	RVK LICENSE	This command is used to revoke the LKF that is in use.
7.3.1 Turning On the Emergency License State Switch	SET LICENSECTRL	This command is used to turn on the emergency license state switch.
Query the status of the emergency license state switch	LST LICENSECTRL	This command is used to query whether the emergency license state switch is turned on.
7.2.1 Querying LKF Information	LST LICENSE	This command is used to query LKF information about an NE.
7.2.2 Querying License Sales Information	DSP LICINFO	This command is used to query the license status, time to enter the grace period, and license configurations of a gNodeB.

To...	Run...	Description
7.1.3 Checking the Consistency Between Licensed Values and Data Configurations	CHK DATA2LIC	This command is used to check whether the data configurations of a gNodeB are consistent with an LKF.
7.3.2 Setting the Switch Controlling Fixed-Term LKFs	SET FIXTERMLICSWITC H	This command is used to control whether fixed-term license control items take effect.
Querying the switch controlling the fixed-term LKF	LST FIXTERMLICSWITC H	This command is used to query the status of the switch controlling fixed-term license control items.
Query license revocation information	DSP RVKLICINFO	Query license revocation information

7.1 Installing LKFs

7.1.1 Prechecking or Installing an LKF

Run the **INS LICENSE** command on the LMT to precheck or install an LKF:

- If the **PREFLG** parameter is set to **YES(Yes)**, this command is used to precheck an LKF. The precheck function also requires that the **FLG** parameter be set to **NO(No)**.
- If the **PREFLG** parameter is set to **NO(No)**, this command is used to download an LKF to an NE and activate the LKF. The LKF can be a commercial, commissioning, or commercially permanent & fixed-term LKF.

7.1.2 Activating an LKF

Run the **ACT LICENSE** command on the LMT to activate an LKF:

- If the **PREFLG** parameter is set to **YES(Yes)**, this command is used to precheck an LKF. The precheck function also requires that the **FLG** parameter be set to **NO(No)**.
- If the **PREFLG** parameter is set to **NO(No)**, this command is used to activate an LKF. The LKF can be a commercial, commissioning, or commercially permanent & fixed-term LKF. If the **FLG** parameter is set to **YES(Yes)**, minor errors that occur during the installation will be ignored and the LKF will be installed forcibly.

7.1.3 Checking the Consistency Between Licensed Values and Data Configurations

Run the **CHK DATA2LIC** command on the LMT to check whether the data configurations of an NE are consistent with an LKF. This LKF can be either a specified file or the LKF that currently takes effect on the NE. If there is any inconsistency, the command output displays the license control items involved and the following information about the corresponding configuration objects: type, name (if involved), attributes (if involved), and quantity (required for resource items).

For a license control item that controls dynamic resources, when the **CHK DATA2LIC** command is executed, the licensed value is checked against the actual consumed value. Such check now applies to the following license control items: NR0S0RRCUL00-RRC Connected User License (NR) and NR0S000SPE00-RE(Resource Element) License (NR).

7.2 Querying License Information

7.2.1 Querying LKF Information

Run the **LST LICENSE** command on the LMT to query LKF information. Information (such as the number, name, activation time, and status) about the LKF on the current NE can be queried.

7.2.2 Querying License Sales Information

Run the **DSP LICINFO** command on the LMT to query the license status, grace period entry time, and license configurations of the gNodeB. The following configurations are included: **Operator Index**, **Operator Name**, **Model**, **Description**, **Allocated**, **Config**, **Actual Used**, **Unit**, and **Expiration Date**.

NOTE

Config indicates the value of a license control item configured on the base station, and it does not indicate the actually consumed value. **Actual Used** indicates the actually consumed value of a license control item after the cell is activated.

7.3 Maintaining LKFs

7.3.1 Turning On the Emergency License State Switch

Run the **SET LICENSECTRL** command on the LMT to turn on the emergency license state switch. After this command is executed, the NE enters the emergency license state. In this state, the NE can serve as many UEs as allowed by the system, without being confined to the licensed values. An NE of an R version can enter the emergency license state up to three times and remain in that state for up to seven days each time.

7.3.2 Setting the Switch Controlling Fixed-Term LKFs

Run the **SET FIXTERMLICSWITCH** command on the LMT to specify whether fixed-term LKFs take effect. The default value is **ON(On)**, indicating that the fixed-term LKFs are effective. This command is effective only for commercially permanent & fixed-term LKFs. When an LKF of any type is reloaded or the base station experiences a cross-R version change, the switch will restore to its default value.

7.4 Possible Issues

7.4.1 Faults Occurred During Installation

- Major errors

If a major error occurs during LKF installation, the installation fails. The following table describes error messages indicating major errors and provides recommended actions.

Error Message	Recommended Action
License file validation failed.	Reapply for an LKF.
The license sequence number is already revoked.	Reapply for an LKF.
The license file does not belong to the product.	- If the software version is correct, apply for and install an LKF with the correct product name. - If the LKF is correct, install it on a gNodeB with the same product name.
License file check failed because the file expires.	Reapply for an LKF.
Days when the commissioning license with ESN set to ANY is allowed to use have elapsed.	Install a commercial LKF, or install a commissioning LKF with ESN not set to ANY .
MCC/MNC in the License File Inconsistent with Primary Operator Configurations.	Apply for and install an LKF with the same MCC and MNC as those of the primary operator on the base station side.
The license file with ESN or version mismatched cannot be activated repetitively.	- If the software version is correct, apply for and install an LKF matching the ESN or version. - If the LKF is correct, install it on a gNodeB with the same ESN or Version.
The ESN indicated in the license file differs from the ESN of the NE.	Apply for and install an LKF with the correct ESN.

Error Message	Recommended Action
The configuration value of the basic control item ^a is greater than the value in the LKF.	Apply for and install an LKF with the basic control item.
The system displays "The value of a license control item in the license file is smaller than its minimum value." and provides a list of involved control items.	Check whether the values of the license control items in the new LKF are smaller than the minimum values. If yes, change the values to be greater than or equal to the minimum values and reapply for a new LKF.
The system displays "The license file has minor errors. The RF module is exiting the deep dormancy state. Please try again after the RF module is woken up. Specifically, you are advised to try again 10 minutes later." or "The RF module is exiting the deep dormancy state. Please try again after the RF module is woken up. Specifically, you are advised to try again 10 minutes later."	The RF module is exiting the deep dormancy state. Wait for 10 minutes and deliver the LKF again.
a: Currently, gNodeB Basic Software is the only basic control item.	

- Minor errors
 - If error messages indicating minor errors are displayed during LKF installation and forced installation is selected, LKFs can be successfully installed.
 - If error messages indicating minor errors are displayed during LKF installation but forced installation is not selected, LKFs can be successfully installed only after the recommended actions are taken.

If a minor error occurs during license installation on a gNodeB and the MAE-Access displays "Major error(Risk involved)" and reminds users that forcible LKF installation will affect services, you are advised to resolve the error and then install the LKF.

Error Message	Recommended Action
The version indicated in the license file differs from the active version of the NE.	• If the software version is correct, apply for and install an LKF with the same Version. • If the LKF is correct, install it on a gNodeB with the same Version. • If neither of these recommended actions is taken or they do not produce the desired result, the gNodeB will enter the 60-day grace period after the LKF is forcibly installed and takes effect.
The validity period of the commissioning license with ESN set to ANY may be insufficient.	If this error is reported during license precheck before the base station is upgraded to a different V/R version (the current base station version is different from the V/R version of the license file), ignore the error. Otherwise, apply for a new license file.
The system displays "The actual value of a license control item is greater than the value in the license file." and provides a list of involved control items.	Check whether the values of the control items in the new LKF are correct. If the values are correct, forcibly install the new LKF. If the values are incorrect, change the values of the control items and reapply for an LKF.
The system displays "The allocated value of the control item is smaller than that in the installed license file."	Check whether the values of the control items in the new LKF are correct. If the values are correct, forcibly install the new LKF. If the values are incorrect, change the values of the control items and reapply for an LKF.
The system displays "The license control item has exceeded the deadline but has not exceeded the grace period." and provides a list of involved control items.	Check whether the expiration dates of the control items in the new LKF are correct. If the dates are correct, forcibly install the new LKF. If the dates are incorrect, change the expiration dates and reapply for an LKF.

Error Message	Recommended Action
The system displays "The license control item has exceeded the grace period." and provides a list of involved control items.	Check whether the expiration dates of the control items in the new LKF are correct. If the dates are correct, forcibly install the new LKF. If the dates are incorrect, change the expiration dates and reapply for an LKF.

The gNodeB configuration takes effect based on the configured values of the control items in an LKF when the LKF is forcibly installed on a gNodeB (with two exceptions). The exceptions are as follows:

- If the licensed value of an item in the LKF is less than Vmin:
For a commercial or commissioning LKF, the licensed value of an item is equal to Vmin. For a commercially permanent & fixed-term LKF:
 - If the licensed value of an item with a commercially permanent date plus the licensed value of the item with a fixed-term date is less than Vmin, then the licensed value of the item with the commercially permanent date is equal to Vmin while the licensed value of the item with the fixed-term date is equal to that in the LKF.
 - If the licensed value of an item with a commercially permanent date plus the licensed value of the item with a fixed-term date is larger than Vmin but the licensed value of the item with the commercially permanent date is less than Vmin, then the licensed value of the item with the commercially permanent date is equal to Vmin while the licensed value of the item with the fixed-term date is equal to that in the LKF.
- If the licensed value of an item in the LKF is greater than Vmax:
For a commercial or commissioning LKF, the licensed value of an item is equal to Vmax.
 - For a fixed-term LKF, if the licensed value is greater than the Vmax, the queried licensed value is the value in the LKF in the default state or Vmax in the non-default state.
 - For a commercially permanent & fixed-term LKF:
 - If the licensed value of an item with a commercially permanent date plus the licensed value of the item with a fixed-term date is greater than Vmax and the licensed value of the item with the commercially permanent date is greater than Vmax, then the licensed value of the item with the commercially permanent date is equal to Vmax while the licensed value of the item with the fixed-term date is equal to 0.
 - If the licensed value of an item with a commercially permanent date plus the licensed value of the item with a fixed-term date is greater than Vmax and the licensed value of the item with the commercially permanent date is less than or equal to Vmax, then the licensed value of the item with the commercially permanent date is equal to that in the LKF while the licensed value of the item with the fixed-term date is equal to Vmax minus the licensed value of the item with the commercially permanent date.

 NOTE

Maximum value (Vmax): indicates the maximum value of a license item supported by an NE

Minimum value (Vmin): indicates the minimum value of a license item supported by an NE

7.4.2 Alarms

Alarm ID	Alarm Name	Alarm Severity
26812	System Dynamic Traffic Exceeding Licensed Limit	Major
26815	Licensed Feature Entering Keep-Alive Period	Major
26816	Licensed Feature Unusable	Major
26817	License on Trial	The alarm severity level is major when the alarm is first reported and changes to critical when the LKF has entered the grace period and the remaining days before expiry is less than or equal to seven days.
26818	No License Running in System	Critical
26819	Data Configuration Exceeding Licensed Limit	Major
26822	Excessive License Usage	Minor
26823	License Feature About to Expire	Minor

 NOTE

The NE automatically checks for alarms with the following IDs from 02:00 to 03:00 every day: 26815, 26816, 26817, 26818, 26819, 26822, and 26823.

- If alarm triggering conditions are not met, the corresponding alarm is not reported.
- When alarm triggering conditions are met and no corresponding alarm has been reported, the alarm is reported.

7.4.3 Events

Event ID	Event Name	Event Severity
26820	License Emergency Status Activated	Warning
26821	License Emergency Status Ceased	Warning

8 License Management Functions of Base Stations

Each NE stores one LKF in its main control board. Each LKF is dedicated to one NE and cannot be shared between NEs except for a commissioning LKF with **ESN** set to **ANY** (which can be installed on any NE). LKFs have one-to-one mapping with ESNs. The ESN of an NE is the ESN of the backplane in its BBU.

- A new LKF is required if the ESN is changed (for example, when the BBU is replaced). After a new LKF is obtained, the LKF can be installed and activated through the MAE-Access or LMT. When the LKF is being installed, the gNodeB checks the consistency between the license control items in the LKF and the configured data.
- If the main control board of an NE is replaced but the ESN remains unchanged, the original LKF can be installed on the new main control board. There is no need to apply for a new LKF.

8.1 LKF Precheck

Before an LKF is installed on an NE, the MAE-Access delivers the LKF to the NE for a precheck to ensure the validity of the LKF. The NE only checks the validity of the LKF, but it does not load the LKF. After the precheck, the NE reports the check results to the MAE-Access, and the MAE-Access generates a result file for operators to determine whether the LKF is valid. For detailed operations, see [6 License Operations on the MAE-Access](#).

8.2 LKF Verification

When an LKF is being installed on an NE, the NE verifies the LKF and takes corresponding actions based on the verification results.

The following situations may occur during LKF installation:

- If the LKF has expired, it fails to be installed.
- If the product name or the signature in the **Footer** segment fails verification, the LKF fails to be installed.

- If the MCC or MNC in the LKF is inconsistent with that configured for the primary operator on the base station side, the LKF fails to be installed.
- If the value of **ESN** is not the actual ESN of the NE, the LKF fails to be installed.
- If the value of **Version** does not match the version of the NE, the installation result indicates the error and the operator can choose to forcibly install the LKF. After forced installation, the NE operates in the grace period state.
- If the value of a license control item (excluding the basic item such as the number of cells per gNodeB) in the LKF is less than the specific configured value, the installation result lists the license control item. The operator can choose to forcibly install the new LKF. If a license control item is not available from the LKF, the corresponding function is license controlled at the minimum value (Vmin, indicating the minimum value of a license item supported by an NE).
- The license control items NR0S0RRCUL00-RRC Connected User License (NR) and NR0S000SPE00-RE(Resource Element) License (NR) can be checked during LKF installation. This function is controlled by the **RRC_OR_RE_NUM_LIC_CHK_SW** option of the **gNBLicenseAlmThld.LicenseChkSw** parameter. When this option is selected, the base station checks the licensed values of license control items in a new LKF to be installed. If the licensed value of a license control item in the new LKF is less than that in the existing LKF, the base station displays "Minor errors occur." and lists such license control items. However, operators can still forcibly install the new LKF. Assume the new LKF is forcibly installed. When the actually used value of such a license control item exceeds its licensed value, the NE reports ALM-26812 System Dynamic Traffic Exceeding Licensed Limit and the configured data beyond the licensed value does not take effect. When this option is deselected, the base station does not check the licensed values of license control items.

8.3 Consistency Check Between Licensed Values and Data Configurations

Consistency check between licensed values and data configurations can be used in the following scenarios:

- LKF installation
A message indicating a minor error is displayed on the LMT and a message indicating a major error is displayed on the MAE-Access if a license control item in an LKF indicates a value less than the corresponding configured data. If the LKF is forcibly installed, the NE reports ALM-26819 Data Configuration Exceeding Licensed Limit and the configured data beyond the licensed value does not take effect. For details, see [7.4.1 Faults Occurred During Installation](#). LKF installation fails if the license control item "gNodeB Basic Software" is unavailable.
- Data configuration
An NE compares data configurations with licensed values when it attempts to make the data configurations take effect, rather than when the data is configured. Data can be written into the database when data configuration constraints are fulfilled. When the NE attempts to make the data

configurations take effect, a failure message is displayed if the NE finds that the configured data is greater than the licensed values. In addition, the NE reports ALM-26819 Data Configuration Exceeding Licensed Limit.

- License maintenance before and after LKF installation

Before a new LKF is installed on an NE, related consistency check can be performed to verify whether the LKF meets the NE configuration requirements. For details about how to perform the check, see [6 License Operations on the MAE-Access](#). Consistency check can also be performed when ALM-26819 Data Configuration Exceeding Licensed Limit is reported because the delivered configuration data exceeds the licensed limits after an LKF is installed on an NE. In this situation, consistency between the LKF and NE data configurations can be checked. For details about how to perform the check, see [6 License Operations on the MAE-Access](#).

8.4 LKF Revocation

A new LKF is required to replace an existing commercial LKF if the related ESN is changed or the licensed capacity is adjusted. Before applying for a new LKF, ensure that the previous commercial LKF has been revoked and a revocation code has been obtained. The descriptions of LKF revocation and revocation codes are as follows:

- LKF revocation is the retrieving of LKFs to ensure that they will not be reused on devices. To revoke an LKF, perform the revocation operations on the MAE-Access client as described in Revoking an LKF in [6 License Operations on the MAE-Access](#), or run the **RVK LICENSE** command on the LMT.
- A revocation code is a character string generated after an LKF is revoked. A license revocation code is proof of proper LKF revocation and a prerequisite to applying for a new LKF.

After the LKF is revoked, a revocation code is returned by the NE. The NE enters the grace period state, retaining its services for up to 60 days.

When an LKF on an NE is revoked, the NE reports ALM-26817 License on Trial.

8.5 License Overload

During system operation, the capacity resource items, such as the UE quantity, may be confined to the licensed limits due to high usage. This license insufficiency affects services that require resources beyond the licensed limits. To promptly inform users that capacity needs expanding, ALM-26812 System Dynamic Traffic Exceeding Licensed Limit is reported when the usage of the capacity resource items exceeds the licensed value.

The overload alarm detection mechanism is as follows:

- When the usage of a capacity resource item exceeds the alarm reporting threshold, the NE reports ALM-26812 System Dynamic Traffic Exceeding Licensed Limit. The alarm reporting threshold is specified by the **gNBLicenseAlmThld.AlarmGenerateThld** parameter.
- When the usage of a capacity resource item falls below the alarm clearance threshold, ALM-26812 System Dynamic Traffic Exceeding Licensed Limit is

cleared. The alarm clearance threshold is **gNBLicenseAlmThld.AlarmGenerateThld** minus 0.1.

An alarm is reported if a dynamic resource item meets the alarm reporting condition at all check points within an alarm detection period (set to 5 minutes by default). The alarm is cleared if the item meets the alarm clearing condition at all check points within an alarm clearance period (set to 5 minutes by default). The alarm detection period and alarm clearance period can also be specified by the **gNBLicenseAlmThld.AlarmChkPeriod** parameter.

8.6 LKF Backup and Restoration

LKFs contain important NE data and they can be backed up and restored using the MAE-Access. If the LKF on an NE must be reloaded due to LKF damage, an NE version rollback, or other problems (for example, a damaged main control board), the LKF can be restored using the MAE-Access. The MAE-Access supports LKF backup and restoration in batches for NEs. LKF backup and restoration using the MAE-Access can be performed through the following methods:

- Automatic backup of LKFs
If an LKF is successfully installed on an NE using the MAE-Access, the LKF is uploaded to the backup directory of the MAE-Access server. If the installation fails, the LKF is not automatically backed up.
- Manual backup of LKFs
If an LKF is installed on an NE using the LMT, the LKF is not automatically backed up on the MAE-Access server. Operators can select a batch of NEs using the MAE-Access client and upload the LKFs of the NEs to the backup directory of the MAE-Access server. For details about how to manually back up LKFs using the MAE-Access, see [6 License Operations on the MAE-Access](#).
- Manual restoration of LKFs
The LKFs can be restored from the backup directory of the MAE-Access. Specifically, the MAE-Access searches its backup directory for the LKF of the NE based on the NE name and ESN. If the LKF is available, the LKF is installed again. If the LKF is unavailable, the NE is not displayed in the list of candidate NEs for LKF restoration and the LKF of the NE cannot be restored. In addition, LKF restoration can be implemented by downloading LKFs from the MAE-Access to the local PC. In this case, restoring LKFs in batches is supported. For details about how to restore LKFs, see [6 License Operations on the MAE-Access](#).

Both automatic and manual LKF backup using the MAE-Access adheres to the following principles:

- Backup LKFs are named in the format of "*NE name_ESN_Backup time*". After backup, the original LKF (if any) is replaced by the new LKF, ensuring that only one LKF maps to an NE.
- During automatic and manual LKF backup, LKFs that are in use and those not in use are both backed up, and they are stored in subdirectories **active** and **inactive** in the backup directory of the MAE-Access server, respectively.

8.7 Invalidating Fixed-term License Items

If an operator determines that all fixed-term feature segments in a commercially permanent & fixed-term LKF are no longer required, the operator can run the **SET FIXTERMLICSWITCH** command to invalidate their license items with fixed expiration dates. Then, the corresponding features become unusable, and ALM-26816 Licensed Feature Unusable or ALM-26815 Licensed Feature Entering Keep-Alive Period will not be reported when a licensed feature expires or the grace period of an expired feature controlled by the license elapses.

This function does not take effect for other types of LKFs.

When an LKF of any type is reloaded or a base station goes through a cross-R version change, the fixed-term license switch will be restored to the default value (**ON**), which indicates that the fixed-term license items are effective.

NOTICE

An operator can perform this function only after determining that all the fixed-term license items in the LKF are no longer required. Otherwise, ongoing services may be affected due to the lack of license items, and ALM-26819 Data Configuration Exceeding Licensed Limit may be reported.

The base station reports ALM-26819 Data Configuration Exceeding Licensed Limit immediately, after the LKF is forcibly installed and thereby the number of license units decreases. In this case, the base station does not forcibly deactivate the activated cells. When cell reactivation is triggered by an upgrade, fault recovery, or configuration, the number of license units is checked again. If the license resources are insufficient, the cells cannot be activated.

9 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *License Control Item Lists*
- *License Management in GBSS Feature Documentation and RAN Feature Documentation*
- *License Management in eRAN Feature Documentation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Operation and Maintenance in iMaster MAE Product Documentation*